

Event rates by group in a field experiment on nitrogen fertilisation in grassland plots

Abstract

Background: A field experiment on nitrogen fertilisation in grassland plots. Methods: The proportion of plots with the study event was compared between the fertilised and unfertilised groups with a Pearson chi-square test of independence without continuity correction. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a field experiment on nitrogen fertilisation in grassland plots. The primary question was whether nitrogen fertilisation is associated with above-ground biomass (g/m^2). The dataset accompanying this report (data.csv) contains one row per plot and is described in the data dictionary.

Methods

Participants and data. Data were collected for 310 plots. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `treatment`.

Exclusion criteria and preprocessing. Only plots aged 18 to 80 years inclusive were included in the analysis. Plots with missing values for biomass were excluded (complete-case analysis).

Statistical analysis. The proportion of plots with the study event was compared between the fertilised and unfertilised groups with a Pearson chi-square test of independence without continuity correction. The risk ratio is reported as the event proportion in the fertilised group divided by that in the unfertilised group. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

Results

Of 135 plots in the fertilised group, 94 experienced the event (proportion 0.696) compared with a proportion of 0.429 among the 163 in the unfertilised group (risk ratio 1.62; chi-square = 21.25, $p < 0.001$).

Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.